SCI-CAL

TolTEC Timestream Calibration
Scientific Basis, Assumptions, and Validation

Scientific contract version 0.1

Rationale revision 0.5 – DRAFT

2026-08-20

A science-team explanation of the calibration chain, its physical meaning, uncertainty boundaries, validity limits, and evidence still required. The separate SCI-CAL engineering contract remains the normative conformance authority.

Scientific owner: Grant Wilson

Executive summary

A TolTEC detector timestream is placed on a top-of-atmosphere, point-source-equivalent flux-density scale by applying one detector-specific calibration factor from the selected observation-specific array property table (APT), whose calibration descends from Beammap calibration evidence, and one atmospheric correction for the target sample. For detector (d), time (t), observation (o), and array (a(d)), the organizing equation is

$$d_d^{\mathrm{cal}}(t) = F_{od}^{\mathrm{APT}}\, C_{a(d),\alpha}[\tau_{225,o}(t), EL_o(t)]\, x_d^{xs}(t). \tag{1}$$

Here $x^{xs} = \Delta f/f_{\mathrm{res}}$ is the dimensionless ordinary `xs` calibration input, F^{APT} is the selected row's `flxscale`, and C_a is the inverse atmospheric transmission for the target observation. The result is a point-source-equivalent, beam-peak-normalized amplitude intended to produce mJy beam^{-1} when the downstream map and filtering response has unit peak, or has been explicitly renormalized to unit peak.

The quantity τ_{225} is dimensionless zenith optical depth measured at 225 GHz, and X is the sample airmass. Their broadband, array-dependent mapping is discussed in Section 4.

The selected `flxscale` is applied exactly once. If an explicitly approved TolProj child transformation has already embodied a pointing-derived correction in the selected child APT, that ancestry is recorded but the correction is not multiplied again. None of `responsivity`, `sens`, a parent `flxscale`, or an opaque `fcf` is an additional absolute signal multiplier.

Intended science model	Equation (1), with one selected detector factor and the target sample's atmospheric correction.
Numerical atmosphere status	Content-bound operator `am12_fixed_djf25_piecewise_linear_los_tau_v1`; support $0 \le \tau_{225} \le 0.25$, $25° \le EL \le 80°$, and $\alpha \in \{-1, 0, 2, 4\}$; fail closed outside.
Implementation status	Not assessed. This rationale neither audits nor certifies a Citlali implementation.
Observational status	Not validated here. Atmosphere fidelity, repeatability, and absolute source recovery require separate evidence.

Measurement-noise estimation is downstream of CAL. CAL should ultimately report array-dependent systematic calibration uncertainty; the required producer mechanisms and covariance model are not yet available. Common terms do not average down as independent detector samples. Formal machine states, identity mechanics, exact requirements, and pathological tests remain in the engineering contract and are traced in the appendices.

Contents

1 What is being calibrated?

SCI-CAL v0.1 is restricted to the ordinary `xs` detector stream. The selected observation-specific APT and its associated calibration lineage identify the source-calibration row used for each target detector. An ambiguous or unsupported match produces no calibrated value for that detector.

Ordinary `xs` is the fitted KID resonant-frequency fractional change $\Delta f / f_{\mathrm{res}}$. It is dimensionless, and increasing `xs` means increasing absorbed optical power. The fitted resonance is expected to match the active probe tone. The stream responds to changes in total optical loading but has no absolute DC sensitivity. SCI-CAL performs no additive baseline subtraction or additional normalization.

> **Scientific-owner decision: Input signal and operating domain**
> Ordinary `xs` has the definition above and is treated as linear over expected operational loadings. A wrong Tune invalidates that premise. Displacement beyond about half a resonance width is a high-noise quality condition, not an alternate calibration observable.

Polarization calibration, auxiliary measured streams, surface-brightness or temperature calibration, extended-source calibration, and integrated photometry do not inherit Equation (1); they are outside v0.1.

2 The physical calibration model and reduction chain

A minimal generative picture separates detector response from atmospheric attenuation. For a point-source-equivalent top-of-atmosphere signal $s_d(t)$, the input can be represented schematically as

$$d_d^{\mathrm{in}}(t) = R_{od} T_{a(d)}(t) s_d(t) + n_d(t), \qquad (2)$$

where R is an abstract detector response in input units per mJy beam^{-1}, T_a is transmission, and n is measurement noise. The central equation is the inverse operation when $F^{\mathrm{APT}} = R^{-1}$ and $C_a = T_a^{-1}$. Here R is not the APT field `responsivity`. This motivates factor direction; the frozen SCI-BEAM contract establishes the source-factor derivation described below.

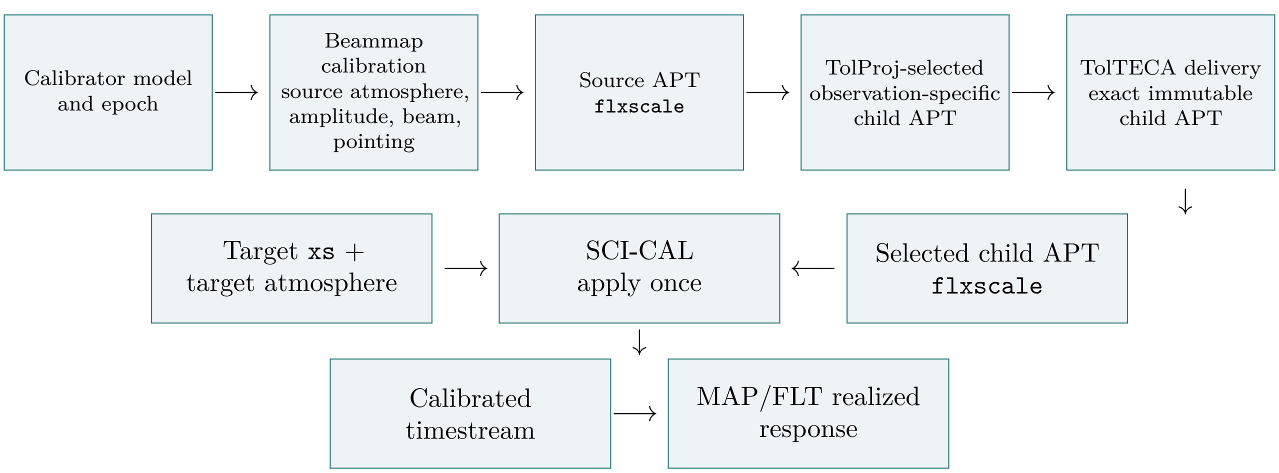

Figure 1: Scientific lineage from calibrator evidence to target-observation calibration. Beammap calibration produces the source APT factor; TolProj selects or creates the observation-specific child APT; TolTECA delivers that exact immutable artifact without changing its value or meaning; SCI-CAL consumes its selected `flxscale` and applies the target atmosphere once.

SCI-CAL lies between upstream definition of the signal, APT, opacity, and airmass and downstream PTC cleaning, filtering, weighting, mapmaking, and inference. The relevant ordering cases are

$$\begin{aligned} L_d[F_d x_d] &= F_d L_d[x_d], && \text{fixed scalar, single-detector linear operator,} \\ P[F\boldsymbol{x}] &\neq F P[\boldsymbol{x}], && \text{detector-mixing operator in general,} \\ L_d[C_d(t)x_d(t)] &\neq C_d(t) L_d[x_d(t)], && \text{sample-dependent atmosphere in general.} \end{aligned} \tag{3}$$

SCI-CAL applies the absolute and target-atmosphere factors before PTC. PTC then owns DC removal and correlated atmospheric/common-mode cleaning. Common-mode removal and PCA mix detectors; a PCA basis may also be data-dependent, so the declared order is scientifically material. Downstream uncertainty or response companions on the calibrated domain must use the same realized multiplier and support.

3 From a calibrator Beammap to the target observation

Beammap analysis measures detector response to a calibrator whose top-of-atmosphere flux model is adopted for the relevant epoch and photometric convention. The frozen SCI-BEAM authority defines

$$F_{sd}^{\text{source}} = \frac{F_{\text{TOA,nom}}}{A_{\text{TOA-equivalent,sd}}}. \tag{4}$$

Its units are $\text{mJy beam}^{-1}/(\Delta f/f)$. Beammap fits the standardized detector response with the declared finite-source-convolved effective-core model. Its generating record identifies the calibrator and epoch, source model, nominal beam, passband/spectrum, source-atmosphere treatment, fit support, acceptance, and available uncertainty/covariance.

Beammap calibration and source-APT production own the calibrator model, the source-observation atmosphere, the Beammap amplitude and beam/template fit, the pointing treatment in source calibration, and the derivation and meaning of source-APT `flxscale`. TolProj owns target-to-source detector association, selection or creation of the observation-specific child APT, and only explicitly approved child transformations. TolTECA owns delivery of the exact immutable selected child artifact and its identity; delivery changes neither the child value nor its producer- and transformer-owned scientific meaning. SCI-CAL consumes that delivered child row as the value boundary, applies its `flxscale` once, and applies target atmosphere. The downstream MAP/FLT response owner owns the realized mapmaker, gridding, kernel, and filter response.

This follows the producer–transformer–delivery–consumer rule: the producer owns a quantity's meaning, a transformer owns only its approved transformation and lineage, delivery preserves the exact selected artifact, and a consumer applies the selected value without reinterpretation. If $F^{\text{child}} = PF^{\text{parent}}$, the right side records ancestry; neither P nor F^{parent} is then applied to the target signal.

Every target detector needs one unique, traceable, observation-valid match to one measured APT row. Exact mechanics remain in the engineering contract; an unmatched or ambiguous detector is excluded rather than calibrated from row position or a plausible default.

Scientific-owner decision: Calibration selection and optional rescale
The default is the temporally closest scientifically accepted Beammap source APT; no universal maximum age or same-Tune, loading, focus, or detector-state condition is imposed. At the reducing scientist's direction, TolProj may create a child APT with one per-array photometric rescale based on a suitable recent ALMA, SMA, planet, or well-calibrated asteroid reference. Its lineage records the source, authorities/epochs, inferred and reference fluxes, spectral convention, factor, and uncertainty when available. Primary-mirror efficiency variation with aberration remains a hypothesis, not an established calibration law.

Table 1: Roles of calibration-related quantities.

Quantity	Physical role	Source	Signal factor?	Correlation or uncertainty scope
`flxscale`	Selected absolute detector scale	Selected observation-specific child APT; derived from source-APT/Beammap calibration	Yes, once	Detector plus shared calibrator ancestry
Pointing correction	Correction in APT ancestry	Beammap source calibration or an explicitly approved child transformation	No, if embodied	Detector/observation correlated
`responsivity`	Relative response or diagnostic	Selected APT lineage	No	Transfer role unresolved
`sens`	Sensitivity or approximate weight	Beammap analysis	No	Detector/cohort; not total error
Target C_a	Inverse transmission for target sample	Opacity, airmass, array operator	Yes, once	Time, observation, and array correlated
Compatibility `fcf`	Declared decomposition of known roles	Compatibility boundary	Not independently	Inherits constituent scopes

4 Atmospheric extinction correction

The LMT water-vapor radiometer supplies τ_{225}. Its raw readings are written into the observation's `tel*.nc` files approximately every five minutes and are interpolated in time to the detector sample grid. Full sample airmass is used with $X_{\mathrm{ref}} = 0$. The retained operator is `am12_fixed_djf25_piecewise_linear_los_tau_v1`, generated from the fixed `LMT_DJF_25.amc` profile.

For the frozen TolTECA-v1 array response $R_a(\nu)$, reference spectrum $S_\alpha \propto \nu^\alpha$, and AM transmission, it evaluates

$$T^{\mathrm{eff}}_{a,\alpha}(\tau, EL) = \frac{\int R_a S_\alpha T_{\mathrm{AM}}\, d\nu}{\int R_a S_\alpha\, d\nu}, \qquad C_{a,\alpha} = 1/T^{\mathrm{eff}}_{a,\alpha}. \tag{5}$$

The supplied energy-weighted throughput is integrated by composite trapezoid on its exact nodes without spectral extrapolation. The authoritative ordinate is $-\ln T^{\mathrm{eff}}$: PCHIP in elevation at each opacity anchor, then piecewise-linear interpolation across opacity anchors. The surface has an analytic unity anchor at zero and no switch at $\tau_{225} = 0.15$.

> **Scientific-owner decision: Authoritative atmosphere operator**
> The exact support is $0 \le \tau_{225} \le 0.25$, $25^\circ \le EL \le 80^\circ$, and $\alpha \in \{-1, 0, 2, 4\}$, with default $\alpha = 0$ and no interpolation or extrapolation in α. The operator fails closed outside support. Its authority commit is `7156881bd1a47e8cece97b8c541a013c93ac03e1`; the contract and node-table digests are recorded in the engineering authority.

CAL assigns one class to the entire observation and never splits it into science and engineering pieces. The nominal 0.15 science and approximately 0.25 engineering limits are flexible experience-based operational guidance, not physical or calibration-error discontinuities. A 0.025 tolerance allows momentary excursions without invalidating the whole observation. This quality tolerance never expands the numerical operator: an individual sample outside support remains unavailable. Policy changes require owner approval.

5 Photometric convention and downstream response

The intended output is a top-of-atmosphere, point-source-equivalent, beam-peak-normalized amplitude. “Beam” denotes the originating Beammap response normalization; it is neither pixel area nor an approved definition of surface brightness or integrated flux.

For a source whose flux density is S mJy, originating unit-peak template P_B, and realized map-maker/gridding/filter response (L),

$$m_{\text{peak}} = \rho_L S \text{ mJy beam}^{-1}, \qquad \rho_L = [LP_B](\mathbf{0}). \tag{6}$$

The literal peak equals S only for $\rho_L = 1$, or after documented renormalization. A valid smoothing kernel can broaden a point source and reduce its peak; retaining the old label without response correction would then misstate flux.

TolTEC is broadband. The per-array reference frequencies used by `toltec_beammap` are 272, 214, and 150 GHz for a1100, a1400, and a2000. Calibration-source spectra are source dependent and commonly represented by $S_\nu \propto \nu^\alpha$. BEAM/`toltec_beammap` supplies one per-array mJy beam^{-1} input to `flxscale`; SCI-CAL applies it without a second source color correction.

Scientific-owner decision: Broadband photometric convention
Target-source color correction is a downstream scientist action. Until that action, the product retains its declared reference-spectrum convention. The atmosphere model uses one array-average passband per array; detector/network passband variation and its uncertainty are presently unavailable.

6 Uncertainty: measurement noise is not total calibration error

Measurement-noise estimation is downstream of CAL; CAL does not create a statistical variance or weight. For fixed calibration, atmosphere, detector association, and response, let $M_j = F_{od}^{\text{APT}} C_{a(d),\alpha}(t)$ and $A = \text{diag}(M_j)$. A downstream covariance on the same support must remain consistent with

$$C_{\text{cal}|\theta} = AC_{\text{in}|\theta}A^{\mathsf{T}}, \qquad \sigma^2_{\text{cal},j} = M_j^2 \sigma^2_{\text{in},j}, \quad w_{\text{cal},j} = w_{\text{in},j}/M_j^2. \tag{7}$$

This is a downstream consistency identity, not a CAL uncertainty product. Missing uncertainty is unavailable, never zero. Calibration-state uncertainty is different. A fractional scale uncertainty (s) common to affected samples contributes

$$C_{\text{common}} = s^2 \boldsymbol{\mu}\boldsymbol{\mu}^{\mathsf{T}}, \tag{8}$$

which does not decrease merely because more identically affected samples are averaged.

CAL should ultimately report an array-dependent systematic calibration error for scientist use. Scale terms are common within each array. WVR state is observation-common across arrays but has an array-dependent response; numerical cross-array covariance is unavailable. Until the source-APT, TolProj-rescale, WVR, and covariance mechanisms exist, total uncertainty and significance remain unavailable, not zero.

7 Validity limits and what a science user sees

Table 2: Science-facing uncertainty budget. "Not established" means the materials available for this draft do not state a present numerical product.

Quantity	Typical correlation scope	In sample weight?	Current status	Interpretation
Measurement noise	Sample/detector, possibly correlated	Downstream	Downstream-owned	Must obey ACA^T on calibrated support
Detector `flxscale` fit	Detector/Beammap cohort	No	Unavailable; BEAM mechanism needed	Calibration systematic
Calibrator scale/model	Observations and/or band	No	Not established	Does not average down
Pointing rescale	Array/observation	No	Child-APT mechanism undefined	Embodied once
Atmosphere, WVR, operator	Observation-common driver; array response	No	Uncertainty unavailable	CAL must ultimately estimate
Passband/source spectrum	Array	No	Variation unavailable	Broadband systematic
Beam/filter response	Product/reduction state	No	Downstream-owned	Needed for literal peak
`sens` approximation	Detector/cohort	Only if declared	Not established	Never total error

Table 3: Meaning of calibration status for science users.

Status	Calibrated value?	Conditional weight?	Interpretation and action
Structurally complete within nominal domain	Yes, once numeric inputs are supported by approved definitions	If supplied	Algebra and inputs complete; atmosphere fidelity and on-sky performance remain separate.
Calibration disabled	No	No calibrated weight	Intentional uncalibrated mode; do not relabel input.
Unsupported stream or unit	No	No	A separate scientific contract is required.
Invalid/missing atmosphere	No	No	Correct the owning input or exclude segment.
Outside operator support	No	No	No extrapolation, clamp, or legacy fallback.
Engineering-only observation	Yes on operator-supported samples	Downstream	Same calibration operator; no science-quality claim.
Conditional uncertainty unavailable	Signal may exist	No	Do not interpret missing uncertainty as zero.

Status	Calibrated value?	Conditional weight?	Interpretation and action
Total uncertainty incomplete	Signal may exist	Conditional only	Report noise separately from systematics.
Response basis incomplete	Signal may exist	Conditional only	Withhold literal peak meaning.

The engineering state `science-qualification-eligible` is rendered here as "structurally complete but not yet scientifically validated."

8 What has to be validated, and what each validation establishes

1. **Structural calibration correctness.** Factor direction and once-only challenges; unit and support checks; unique detector to APT association; zero opacity, node, seam, monotonicity, and no extrapolation behavior; downstream covariance consistency; and response bookkeeping. Software-oriented permutation, duplicate-key, replay, sentinel, and application-count tests remain in the engineering contract.
2. **Atmosphere-model representation fidelity.** Compare the finite operator with its named higher-fidelity generating calculation over each array's declared opacity, elevation, spectral, and passband domain. Report the actual result; the prior approximately one-percent figure is only a benchmark.
3. **Beammap closure.** Reduce calibrated Beammaps to source APTs, validate them with `toltec_beammap`, enter accepted APTs into `apt_library`, associate them through TolProj, and reduce each generating Beammap through the ordinary science path. Recovery of the input calibrator flux establishes APT–TolProj–CAL–downstream closure, not independent absolute calibration accuracy.
4. **Associated-pointing transfer.** Apply each accepted Beammap APT to its associated pointing observation through the same TolProj and science path. When the pointing source has an adequate independent ALMA, SMA, planet, or well-calibrated asteroid flux, this is a stronger calibration-transfer test. Repeat over the observations actually available and report separately for every array.

The record identifies observations, source and child APTs, TolProj associations, flux references, arrays, atmosphere, recovered/input ratios or residuals, repeatability, absolute recovery, and observed condition dependence. No arbitrary combinatorial matrix or minimum sample size is required. The one-percent, five-percent, and five-to-ten-percent figures are reporting benchmarks, not automatic pass/fail ceilings. The task is to report what the system achieves honestly and as accurately as possible; final scientific acceptance is an explicit owner decision.

9 Scientific owner decisions and remaining unavailable products

Table 4: Decision register for science-rationale revision r0.5.

ID	Issue	Owner disposition	Remaining limitation
Q01	Physical `xs` definition	Decided: dimensionless $\Delta f/f_{\mathrm{res}}$, positive optical-power direction, no CAL baseline	Wrong Tune invalidates; half-width excursion is high-noise state

ID	Issue	Owner disposition	Remaining limitation
Q02	Baseline and ordering	Decided: CAL before PTC; PTC owns DC/common-mode removal	Downstream transfer remains separately measured
Q03	`flxscale` derivation	Decided by frozen SCI-BEAM authority	Producer uncertainty presently unavailable
Q04	Calibrator-target transfer	Decided: closest accepted Beammap; optional scientist-directed per-array TolProj rescale	Rescale uncertainty mechanism unavailable
Q05	Broadband convention	Decided: 272/214/150 GHz, source-dependent spectrum, downstream target color correction	Detector/network passband variation unavailable
Q06	Atmosphere operator	Decided: content-bound WVR/AM/passband operator	WVR/model uncertainty unavailable
Q07	Opacity/observation policy	Decided: one CAL class, 0.025 excursion tolerance, owner changes policy	Operational guidance is not a performance claim
Q08	Numerical uncertainty	Decided ownership and correlation scope	Total uncertainty/significance unavailable
Q09	Criteria for validated science use	Decided closure-and-transfer workflow and reporting fields	Final acceptance remains an owner decision

References

[1] *SCI-CAL Detector Calibration, Atmospheric Extinction, and Signal Transfer Scope Brief*, owner-approved 2026-08-16.
[2] *SCI-CAL Independent Scientific Core*, with v0.1 supersession cover.
[3] *SCI-CAL Sanitized Conventions and Ownership*, owner-approved 2026-08-16.
[4] *TolTEC v1 Modeled Array Passband Authority Manifest*, bounded reference with unresolved semantics.
[5] *SCI-CAL Scientific Owner Decisions for r0.5/r0.4*, approved 2026-08-20.
[6] *SCI-CAL Atmosphere Operator Machine Contract*, authority commit `7156881bd1a47e8cece97b8c541a013c93ac03e1`.

The sanitized conventions/ownership reference records the lineage of its approved extract. No broader source in that lineage was independently opened or used as an author input. The r0.5 owner-decision record supplies the approved `xs`, pipeline, transfer, photometric, quality, uncertainty, and validation dispositions. The content-bound machine contract supplies the exact numerical atmosphere authority; the frozen SCI-BEAM contract remains the source-factor authority.

A Notation and formal derivations

Table 5: Notation retained for formal traceability.

Symbol	Meaning
(o,d,t,a(d))	Observation, detector, sample time, detector array
$d^{\mathrm{in}}, d^{\mathrm{cal}}$	Upstream-defined input and calibrated output
F_{od}^{APT}	Selected row's oriented absolute `flxscale`
τ_{225}, X, ℓ	Zenith opacity, airmass, line-of-sight coordinate
$\lambda_{a,\alpha}, T_{a,\alpha}, C_{a,\alpha}$	Interpolated log-transmission ordinate, transmission, correction
M_j, A	Realized scalar multiplier and diagonal operator
$C_{\mathrm{in}}, C_{\mathrm{cal}}, C_\theta$	Conditional input/output and nuisance covariance
P_B, L, ρ_L	Beammap template, downstream response, peak response

The owner-declared CAL input has no additive operation:

$$d_j^{\mathrm{in}} = x_j^{xs}. \tag{9}$$

At each opacity anchor the authoritative $\lambda = -\ln T^{\mathrm{eff}}$ surface uses PCHIP in elevation. Between opacity anchors,

$$u = \frac{\tau - \tau_k}{\tau_{k+1} - \tau_k}, \tag{10}$$

$$\lambda_{a,\alpha}(\tau, EL) = (1-u)\lambda_{a,\alpha}(\tau_k, EL) + u\lambda_{a,\alpha}(\tau_{k+1}, EL), \tag{11}$$

$$(T_{a,\alpha}, C_{a,\alpha}) = (e^{-\lambda_{a,\alpha}}, e^{\lambda_{a,\alpha}}). \tag{12}$$

The declared log-transmission ordinate is interpolated before exponentiation. Exact nodes are recovered, seams continuous, $T_a(0) = C_a(0) = 1$, transmission decreases, correction increases, and extrapolation is unauthorized.

For fixed state θ,

$$\mathbb{E}[\boldsymbol{d}^{\mathrm{cal}} \mid \theta] = A\mathbb{E}[\boldsymbol{d}^{\mathrm{in}} \mid \theta], \qquad C_{\mathrm{cal}|\theta} = AC_{\mathrm{in}|\theta}A^{\mathsf{T}}. \tag{13}$$

For small nuisance uncertainty with $H_{jr} = \partial \ln |M_j| / \partial \theta_r$,

$$C_{\mathrm{nuis}} \simeq \mathrm{diag}(\boldsymbol{\mu}) H C_\theta H^{\mathsf{T}} \, \mathrm{diag}(\boldsymbol{\mu}). \tag{14}$$

Shared-data calibration requires material cross covariances. Large, asymmetric, seam-spanning, or discrete uncertainty requires nonlinear samples, a joint posterior, or bounded sensitivity study.

B Exact machine states and formal conformance routing

The engineering authority distinguishes requested, effective, observation-resolved, and realized states and valid signal. Its exact named states include:

- `disabled/uncalibrated`;
- `outside-supported-calibration`;
- `engineering-only`; and

- `science-qualification-eligible`.

It separately records conditional-uncertainty availability, total-uncertainty completeness, and response completeness. Table 3 translates these for users; it does not replace them. Appendix D maps every formal item to this rationale and its continuing engineering home.

C Provenance and document authority

The approved input content digests are:

- Scope Brief: `a3d6332dcc98bfbb638a45aea9be830edca693a4fe7e65d831b5880603c16578`;
- independent core: `106755520b048f601bc60fd04e7b6020e6fa470480ac3105fa7ba269c730a4fe`;
- passband manifest: `2756908181cc466550399ec0a869e6671de7912bd3a935f9aeebf63e3e826617`;
- conventions/ownership: `7e9a630fd183ca04bc3d8bbd21b5e801776b9aad7bd084d48fa7f2c572766520`.

The engineering contract remains normative for requirements 001–050, assumptions 001–011, and edge predictions 001–030. Rationale revision r0.5 changes genre and presentation, not the governed v0.1 algebra.

D Crosswalk to the engineering contract

The engineering contract remains normative. This compact map locates every formal assumption, requirement, and edge prediction without reproducing its full audit-oriented text in the science narrative.

Table 6: Assumption routing.

Formal IDs	Scientist-facing location	Continuing engineering home
`SCI-CAL-ASM-001`, `SCI-CAL-ASM-002`	Sections 1–2; Q01–Q02	Approved input support, ordinary channel, and binding
`SCI-CAL-ASM-003`	Section 3; Figure 1; Q03–Q04	Source APT, TolProj-selected child APT, immutable TolTECA delivery, and association
`SCI-CAL-ASM-004`	Section 1; Appendix A	No-additive-operation CAL input convention
`SCI-CAL-ASM-005`	Sections 2–3; Q03–Q05	Complete factor lineage, direction, plane, and once-only use
`SCI-CAL-ASM-006`	Section 4; Q06–Q07	WVR inputs, time support, operator, and observation policy
`SCI-CAL-ASM-007`	Section 6	Downstream conditional-state covariance boundary
`SCI-CAL-ASM-008`	Section 5	Originating and downstream response ownership
`SCI-CAL-ASM-009`	Sections 4, 7; Q07	Whole-observation quality class
`SCI-CAL-ASM-010`	Sections 4–5; Q05	Passband limitations and photometric compatibility
`SCI-CAL-ASM-011`	Sections 4, 9; Q06	Content-bound numeric operator authority

Table 7: Requirement routing.

Formal IDs	Scientist-facing location	Continuing engineering home
SCI-CAL-REQ-001, SCI-CAL-REQ-002, SCI-CAL-REQ-003, SCI-CAL-REQ-004, SCI-CAL-REQ-005	Sections 1, 7; Q01–Q09	Scope, typed state, authority snapshot, validity, and one-way resolution
SCI-CAL-REQ-006, SCI-CAL-REQ-007, SCI-CAL-REQ-008, SCI-CAL-REQ-009, SCI-CAL-REQ-010, SCI-CAL-REQ-011, SCI-CAL-REQ-012	Section 3; Figure 1	Acquisition/source/child identity, binding, association, delivery, and lifetime
SCI-CAL-REQ-013, SCI-CAL-REQ-014, SCI-CAL-REQ-015, SCI-CAL-REQ-016	Sections 2–3; Q03–Q04	SCI-BEAM generating record, optional child rescale, and once-only multiplier
SCI-CAL-REQ-017, SCI-CAL-REQ-018, SCI-CAL-REQ-019, SCI-CAL-REQ-020	Section 3; Table 1	Relative roles, compatibility, producer/transformer/delivery/consumer lineage
SCI-CAL-REQ-021, SCI-CAL-REQ-022, SCI-CAL-REQ-023, SCI-CAL-REQ-024, SCI-CAL-REQ-025, SCI-CAL-REQ-026	Section 4; Appendix A	WVR meaning, temporal/operator support, and invariants
SCI-CAL-REQ-027, SCI-CAL-REQ-028, SCI-CAL-REQ-029, SCI-CAL-REQ-030, SCI-CAL-REQ-031	Sections 4, 7; Q05–Q07	Whole-observation policy, sample support, passband, and spectrum
SCI-CAL-REQ-032, SCI-CAL-REQ-033, SCI-CAL-REQ-034	Section 6	Downstream covariance/weight boundary and unavailable state
SCI-CAL-REQ-035, SCI-CAL-REQ-036, SCI-CAL-REQ-037, SCI-CAL-REQ-038, SCI-CAL-REQ-039	Sections 2, 6; Appendix A; Q08	Systematic scopes, unavailable covariance, and companions
SCI-CAL-REQ-040, SCI-CAL-REQ-041, SCI-CAL-REQ-042, SCI-CAL-REQ-043, SCI-CAL-REQ-044	Sections 3, 5	Producer-owned beam/template basis, realized response, and unit exclusions
SCI-CAL-REQ-045, SCI-CAL-REQ-046, SCI-CAL-REQ-047, SCI-CAL-REQ-048	Sections 3, 7; Appendix B; Q01–Q09	Claim-specific states, reasons, complete lineage, and product links
SCI-CAL-REQ-049, SCI-CAL-REQ-050	Sections 5, 8; Q09	Separate closure, transfer, fidelity, performance, and acceptance claims

Table 8: Edge-prediction routing.

Formal IDs	Scientist-facing location	Continuing engineering home
SCI-CAL-EDGE-001, SCI-CAL-EDGE-002, SCI-CAL-EDGE-003, SCI-CAL-EDGE-004, SCI-CAL-EDGE-005	Section 4; Appendix A	Zero, monotonicity, zenith, exact-node, and seam checks
SCI-CAL-EDGE-006, SCI-CAL-EDGE-007, SCI-CAL-EDGE-008, SCI-CAL-EDGE-009, SCI-CAL-EDGE-010	Sections 4, 7	Domain, temporal bracketing, engineering and outside-support states
SCI-CAL-EDGE-011, SCI-CAL-EDGE-012, SCI-CAL-EDGE-013	Section 3	Unity, omitted/duplicate/inverted, corrected-APT challenges
SCI-CAL-EDGE-014, SCI-CAL-EDGE-015, SCI-CAL-EDGE-016, SCI-CAL-EDGE-017, SCI-CAL-EDGE-018, SCI-CAL-EDGE-019	Section 3	Row/order/network, missing/duplicate, association/design tests
SCI-CAL-EDGE-020, SCI-CAL-EDGE-021, SCI-CAL-EDGE-022, SCI-CAL-EDGE-023, SCI-CAL-EDGE-024	Section 6	Downstream consistency, unavailable, common, and nonlinear uncertainty tests
SCI-CAL-EDGE-025, SCI-CAL-EDGE-026, SCI-CAL-EDGE-027	Section 5	Unit-peak source, response-changing kernel, beam approximation
SCI-CAL-EDGE-028	Sections 1, 7	Unsupported stream or target
SCI-CAL-EDGE-029	Section 1; Appendix A	Zero signal and no-additive-operation convention
SCI-CAL-EDGE-030	Sections 3, 7	Observation-order state replay